

TRANSACTION HOMEWORK - 24125076

SQL SERVER's default isolation level is READ COMMITTED. While this prevents dirty reads, it does not protect against non-repeatable reads and phantom reads.

Scenerio 1: Non-repeatable Read

We execute `sp_get_department_payroll_summary` and `sp_adjust_instructor_salary` simultaneously as below.

```
EXEC sp_get_department_payroll_summary_with_delay 'AI';
```

```
EXEC sp_adjust_instructor_salary 'I009', 2000;
```

```
EXEC sp_get_department_payroll_summary_with_delay 'AI';
```

Result: When Session 1 performs its second read, the `TotalSalary` will have magically changed because the underlying data was altered and committed by Session 2. This is a non-repeatable read.

The image displays two screenshots of the SQL Server Enterprise Manager interface, showing the results of a query executed in Session 1. The query is `EXEC sp_get_department_payroll_summary_with_delay 'AI'`.

First Screenshot: The query results show a single row for the department 'Artificial Intelligence' with 3 instructors and a total salary of 6,800. The status bar indicates '1 row(s) fetched - 6.04s, on 2026-08-14 at 12:23:48'.

Second Screenshot: The query results show a single row for the department 'Artificial Intelligence' with 3 instructors and a total salary of 7,300. The status bar indicates '1 row(s) fetched'. This change in the total salary from 6,800 to 7,300 demonstrates a non-repeatable read, where the data has been modified by another session (Session 2) between the two reads of Session 1.

Scenerio 2: Phantom Read

We execute `sp_get_department_payroll_summary` and `sp_adjust_instructor_salary` simultaneously.

```
EXEC sp_get_department_payroll_summary_with_delay 'AI';
```

```
EXEC sp_add_new_instructor 'I013', 'Tran Thien Phuc', '0123456789', 'AI', 20000;
```

```
EXEC sp_get_department_payroll_summary_with_delay 'AI';
```

Result: If Session 1 is running under the default READ COMMITTED level, it might calculate the `NumberOfInstructors` and `TotalSalary` based on the old data, completely missing the newly inserted instructor. If Session 1 were to run its query again within the same transaction, a new "phantom" row would appear.

Results 1 Results 1 (2) Statistics 1

EXEC sp_get_department_payroll_summary_with_c | Data filter is not supported

	A-Z department_name	123 NumberOfInstructors	123 TotalSalary
1	Artificial Intelligence	3	7,300

Refresh Save Cancel Export data 200 1 1 row(s) fetched - 6.027s, on 2026-08-14 at 12:24:57

Results 1 Results 1 (2) Statistics 1

EXEC sp_get_department_payroll_summary_with_c | Data filter is not supported

	A-Z department_name	123 NumberOfInstructors	123 TotalSalary
1	Artificial Intelligence	4	27,300

Refresh Save Cancel Export data 200 1 1 row(s) fetched

Solution: To prevent non-repeatable reads and phantom reads, the isolation level for the payroll summary transaction must be elevated to **SERIALIZABLE**. This ensures transactions run sequentially.